

3D SLIDING PUZZLE ENGINE

CREATE. PLAY. SOLVE.



Fully 3D

Built for depth and realism



Customizable

Create your own puzzles



Easy to Integrate

Clean and modular code



Optimized

Smooth performance



USER MANUAL

Table of Contents

1. List of Features	5
2. Quick-Start	6
3. Create & Assign a Puzzle	7
4. Creating Content With Scriptable Objects	8
4.1 Puzzle (SlidingPuzzle > Puzzle)	
4.2 Puzzle Slice Difficulty (SlidingPuzzle > PuzzleSliceDifficulty)	
4.3 Puzzle Scramble Difficulty (SlidingPuzzle > PuzzleScrambleDifficulty)	
4.4 Puzzle Save (SlidingPuzzle > PuzzleSave)	
5. Puzzle Manager	10
5.1 Scene References	
5.2 Puzzle Settings	
5.3 Puzzle Data	
5.4 Events	
5.5 Audio	
5.6 Live Stats (Play Mode Only)	
6. Public API	13
6.1 Puzzle Flow	
6.2 Tile Display	
6.3 Information	
7. Save System	15
7.1 How Saves Are Triggered	
7.2 What Is Saved	
7.3 Setting Save Interval to 0	
7.4 Multiple Puzzle Managers	
8. Per-Puzzle Difficulty Override	16

9. Input.....	17
10. Tile Movement.....	18
10.1 Single-Tile Move	
10.2 Multi-Tile Move	
10.3 Animation Blocking	
11. Working With Events	19
11.1 Win Event Common Uses	
11.2 Lose Event Common Uses	
11.3 Implementing a Timer	
12. Customizing the Tile Prefab	20
13. Multiple Puzzle Managers in One Scene	21
14. Frequently Asked Questions.....	22
15. Quick Reference Card.....	23

Thank you for purchasing the 3D Sliding Puzzle Engine package.

This engine lets you easily add a sliding puzzle to your project, whether for a full puzzle game with multiple variations and difficulties or as a mini-game within a larger adventure.

This manual walks through every setting, explains how the save system works, and shows you how to wire up events so the puzzle fits naturally into your project.

1. List of Features

- **Easy Setup:** Drag the puzzle prefab into your scene and you're ready to go. Minimal configuration required to get up and running.
- **Scriptable Objects:** Puzzles, slice difficulties, and scramble difficulties are all defined as Scriptable Objects, making it simple to create and tweak content without touching any code.
- **Flexible Slice Options:** Configure puzzles with any row and column combination, including both even (3x3, 5x5) and uneven grids (3x5, 4x2).
- **Per-Puzzle Difficulty Overrides:** Individual puzzles can override the global slice and scramble difficulty settings, giving you fine-grained control over each puzzle in your collection.
- **Adjustable Scrambling Difficulty:** Control how thoroughly each puzzle is scrambled before the player sees it.
- **Re-Scramble Support:** Restart the current puzzle with a fresh scramble without needing to load a new one.
- **Customizable Puzzle Tiles:** Swap in your own 3D models for the puzzle tiles.
- **Tile Movement Speed & SFX:** Tune the tile slide speed, movement sound, and win sound independently.
- **Input Support:** Works with mouse input on desktop and touch on mobile via Unity's New Input System.
- **Simultaneous Tile Movement:** Tapping a tile that isn't directly adjacent to the empty slot slides the entire row or column into place at once.
- **Optional Tile Numbers:** Toggle tile numbers on or off at any time, both in the editor and at runtime.
- **Puzzle Save System:** Progress is saved automatically at a configurable move interval and whenever the app loses focus, so players can always pick up where they left off.
- **Puzzle Order Customization:** Serve puzzles in sequential or random order, with smart random cycling so the same puzzle doesn't repeat until the full set has been seen.
- **Win & Lose Events:** Trigger custom Unity events on both win and lose outcomes, making it straightforward to hook in timers, score tracking, or scene transitions.
- **Multiple Simultaneous Instances:** Each Puzzle Manager component is fully self-contained, so you can run several puzzles side by side in the same scene without any conflicts.
- **Custom Inspector:** A color-coded, collapsible inspector with inline validation warnings and a live move counter keeps the editor experience clean and easy to navigate.

2. Quick-Start



Drag the **PuzzleBox** prefab into your scene and you're ready to go, just hit Play and try it out!

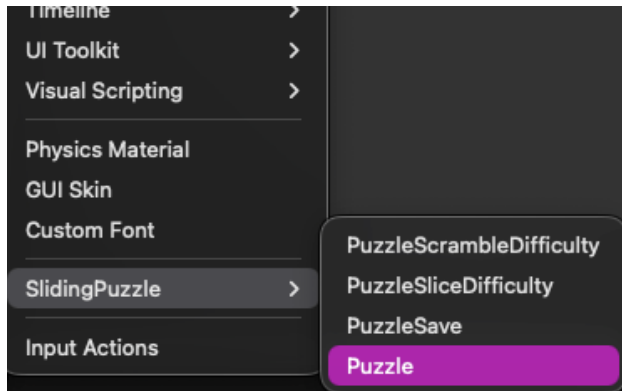
The **PuzzleBox** prefab, is a container that includes a 3D model of the puzzle box with the **PuzzleManager** script attached to it.

Tip

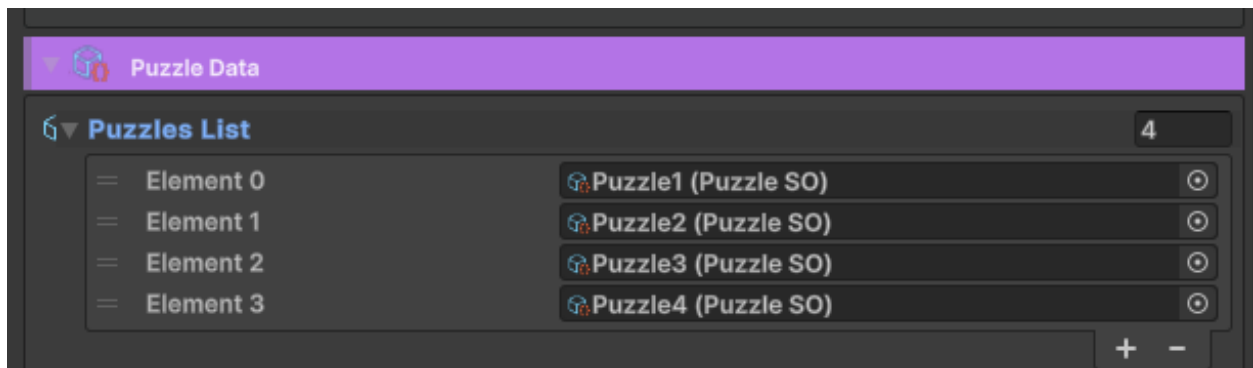
You can support multiple independent puzzles in the same scene by placing more than one **PuzzleBox** object. Each instance tracks its own state and does not interfere with the others.

3. Create & Assign a Puzzle

- **Create a Puzzle asset.** In the Project window, right-click and choose [SlidingPuzzle](#) > [Puzzle](#). Assign your [Texture2D](#) image to the [Puzzle Image](#) field.



- **Assign the puzzle.** Select the [PuzzleManager](#) object, open the Puzzle Data section, and add your new Puzzle asset to the [Puzzles List](#).



- **Hit Play.** The puzzle loads and scrambles automatically if [Auto Start Puzzle](#) is enabled (it is by default).

4. Creating Content With Scriptable Objects

All puzzle content is defined through Scriptable Objects. Right-click anywhere in the Project window to create them.

4.1 Puzzle (SlidingPuzzle > Puzzle)

Setting	Type / Values	Description
Puzzle Image	Texture2D	The image that will be sliced into tiles. Any resolution works; the engine scales the tiles to fit.
Override Global Puzzle Difficulty	Toggle	When enabled, this specific puzzle uses its own slice and scramble difficulty instead of the global settings on the Puzzle Manager.
Puzzle Slice Difficulty	SO reference	Visible only when the override toggle is on. Assign a Puzzle Slice Difficulty asset to control the grid size for this puzzle.
Puzzle Scramble Difficulty	SO reference	Visible only when the override toggle is on. Assign a Puzzle Scramble Difficulty asset to control how thoroughly this puzzle is scrambled.

4.2 Puzzle Slice Difficulty (SlidingPuzzle > PuzzleSliceDifficulty)

Setting	Type / Values	Description
Difficulty Name	string	A display label for this difficulty (e.g. "Easy", "Hard").
Slices Per Row	int	How many columns the puzzle is divided into. Use any positive integer; even and uneven grids are both supported.
Slices Per Column	int	How many rows the puzzle is divided into.

4.3 Puzzle Scramble Difficulty (SlidingPuzzle > PuzzleScrambleDifficulty)

Setting	Type / Values	Description
Scramble Count	int	How many random tile moves are made when scrambling the puzzle. Higher values produce harder starting positions.

4.4 Puzzle Save (SlidingPuzzle > PuzzleSave)

The save asset is created once and assigned to the [Save Slot](#) field on the Puzzle Manager. It stores the full puzzle state to disk automatically. You do not need to call any save functions manually.

5. Puzzle Manager

The Puzzle Manager component is the central controller. Select the **PuzzleBox** object in your scene to see its custom inspector, which is divided into six color-coded sections.

5.1 Scene References

Setting	Type / Values	Description
Puzzle Tile Prefab	Transform	The 3D prefab instantiated for each puzzle tile. Swap this out to use your own custom tile model.
Puzzle Container	Transform	Parent object that holds all the spawned tiles. The engine enables and disables this object when showing or hiding the active puzzle.
Puzzle Complete Image	Transform	An optional flat object (e.g. a quad) whose material texture is set to the solved puzzle image. It is shown automatically when the puzzle is won and can also be shown mid-game as a hint.

5.2 Puzzle Settings

Setting	Type / Values	Description
Auto Start Puzzle	Toggle	When enabled, the puzzle starts automatically when the scene loads. Disable this if you want to call <code>NewPuzzleAction()</code> from your own UI.
Puzzle Start Animation	Toggle	Plays a tile fly-in animation each time a new puzzle is created.
Puzzle Start Mode	StartNewPuzzle / LoadSave	Determines what happens at startup. StartNewPuzzle always creates a fresh puzzle. LoadSave loads the last saved state if one exists, or falls back to a new puzzle.

Setting	Type / Values	Description
Puzzles Order	Sorted / Random	Controls the order puzzles are served. Sorted goes through the Puzzles List in order. Random picks a different puzzle each time, cycling through the full list before repeating any.
Puzzle Slice Difficulty	SO reference	Global slice difficulty used for any puzzle that does not have its own override.
Puzzle Scramble Difficulty	SO reference	Global scramble difficulty used for any puzzle that does not have its own override.
Puzzle Tile Movement Speed	float (seconds)	How long (in seconds) a tile takes to slide into its new position. Lower values feel snappier; higher values feel smoother.
Show Tile Number	Toggle	Shows or hides the numbered label on each tile. Can also be toggled at runtime by calling <code>ShowTileNumberAction()</code> .
Save Interval Moves	int	How many player moves between automatic saves. Set to 0 to disable interval saves and only save when the app loses focus. The save also triggers immediately on focus loss regardless of this value.

5.3 Puzzle Data

Setting	Type / Values	Description
Puzzles List	List<PuzzleSO>	All Puzzle assets this manager can serve. Add as many as you like. The Puzzles Order setting controls how they are cycled through.
Save Slot	PuzzleSaveSO	The save asset used to persist and restore game state. One save asset can only track one active puzzle at a time. If you have multiple Puzzle Manager instances, give each one its own save asset.

5.4 Events

Setting	Type / Values	Description
Win Function	UnityEvent	Fired when the player places all tiles in the correct position. Use this to show a score screen, play an animation, or advance to the next level.
Lose Function	UnityEvent	Fired when Lose() is called programmatically. The engine does not call this automatically — it is there for you to trigger from your own timer or conditions.

Note

The win condition is checked automatically after every tile move. You never need to check it

5.5 Audio

Setting	Type / Values	Description
Tile Move SFX	AudioClip	Played every time one or more tiles slide into a new position.
Win SFX	AudioClip	Played when the puzzle is solved.

Tip

Both audio clips are optional. If neither is assigned, the puzzle runs silently and no errors are shown.

5.6 Live Stats (Play Mode Only)

The Live Stats section of the inspector shows the current move count in real time while the game is running. This is read-only and it is useful for debugging or monitoring without needing to add UI yourself.

6. Public API

The following methods can be called from your own scripts or connected to UI buttons via Unity Events.

6.1 Puzzle Flow

Setting	Type / Values	Description
NewPuzzleAction(bool forceNewPuzzle = false)	method	Starts or loads a puzzle. Pass true to always start a brand new puzzle and discard any existing save. Pass false (the default) to respect the Puzzle Start Mode setting.
ReScrambleAction()	method	Re-scrambles the current puzzle image without loading a new one. Useful for a Retry button.
Win()	method	Manually triggers the win state. The complete image is revealed, the Win Function event fires, and the Win SFX plays.
Lose()	method	Manually triggers the lose state and fires the Lose Function event. Call this from your own timer or game logic.

6.2 Tile Display

Setting	Type / Values	Description
ShowTileNumberAction()	method	Toggles tile numbers on and off at runtime.
ShowCompletePuzzleImage(bool isAnimated)	method	Shows the reference image and hides the tile grid. Pass true for a fade-in animation. Works both mid-game (as a hint) and on win.
HideCompletePuzzleImage()	method	Hides the reference image and restores the tile grid. Only has an effect while the puzzle is still in progress.

6.3 Information

Setting	Type / Values	Description
GetCurrentPuzzleIndex()	int	Returns the index of the currently active puzzle within the Puzzles List.
numberOfMoves	int	Public field. Read this at any time to display or evaluate how many moves the player has made on the current puzzle.
currentPuzzleImage	Texture2D	Public field. The texture of the puzzle currently in play.

7. Save System

The save system is automatic. Once a Puzzle Save asset is assigned to the Save Slot field, the engine takes care of writing and reading save data without any extra code from you.

7.1 How Saves Are Triggered

- **On an interval:** After every [Save Interval Moves](#) player moves, the current state is written to disk.
- **On focus loss:** Whenever the application loses focus (the player alt-tabs, minimizes, or switches to another app), the state is saved immediately, regardless of the move interval.
- **Scramble moves are excluded:** Moves made while scrambling the puzzle at startup are never counted or saved, so the move counter always reflects actual player moves only.

7.2 What Is Saved

- The name of the active Puzzle asset.
- The active slice difficulty (name, rows, and columns).
- The world position of every tile.
- The original and current grid position of every tile.
- The total number of moves made.

7.3 Setting Save Interval to 0

Setting [Save Interval Moves](#) to 0 disables interval saves. The game state is then only written when the app loses focus. This is useful if you want to avoid frequent disk writes on lower-end devices.

7.4 Multiple Puzzle Managers

If your scene contains more than one Puzzle Manager, each one must have its own dedicated Puzzle Save asset. Two managers sharing the same save asset would overwrite each other's state.

8. Per-Puzzle Difficulty Override

By default, every puzzle uses the global Puzzle Slice Difficulty and Puzzle Scramble Difficulty set on the Puzzle Manager. You can override both for any individual puzzle.

- **Select a Puzzle asset** in the Project window.
- **Enable** the [Override Global Puzzle Difficulty Settings](#) toggle.
- **Assign** the Puzzle Slice Difficulty and Puzzle Scramble Difficulty assets you want this puzzle to use.

When this puzzle is loaded, those specific difficulties will be used instead of the global ones. Puzzles with the override disabled always use whatever is currently set on the Puzzle Manager.

Tip

This is handy for a progressive puzzle game where early puzzles use a 3x3 grid and later ones step up to 5x5 or 4x6 — each puzzle carries its own grid settings and the global values act as a safe default.

9. Input

Input handling uses Unity's New Input System via [Pointer.current](#), which unifies mouse and touch input under a single API.

- **Desktop:** A left-click ray is cast from the camera. If it hits a Collider tagged [Tile](#), that tile is tapped.
- **Mobile:** The first active touch is treated exactly like a mouse press. No extra configuration is needed.

Note

The engine uses Enhanced Touch Support from the New Input System. This is enabled and disabled automatically by the Puzzle Manager; you do not need to manage it yourself.

Tiles must have their Collider tagged [Tile](#) for input to register. The included tile prefab already has this tag. If you replace the tile prefab with a custom model, make sure its collider carries the same tag.

10. Tile Movement

10.1 Single-Tile Move

Tapping a tile directly adjacent to the empty slot (distance of 1 in grid units) moves just that tile into the empty space.

10.2 Multi-Tile Move

Tapping a tile that shares the same row or column as the empty slot but is not directly adjacent causes every tile between the tapped tile and the empty slot to slide simultaneously. The whole group moves as one action and counts as a single move.

10.3 Animation Blocking

If a tile is still animating when another tap is registered, the new tap is ignored. This prevents queuing up moves faster than the animation can play and keeps the move counter accurate.

11. Working With Events

11.1 Win Event Common Uses

- Show a score or results panel.
- Start an animation.
- Unlock the next puzzle or trigger a scene transition.
- Award currency or achievement badges.

11.2 Lose Event Common Uses

The engine never calls `Lose()` on its own. You call it from your own script — for example, when a countdown timer runs out.

- Show a game-over screen.
- Display a replay prompt.
- Log an analytics event.

11.3 Implementing a Timer

A common pattern is to start a countdown when the puzzle begins and call `puzzleManager.Lose()` when time runs out. Wire your timer UI to the [Win Function](#) event to stop the countdown when the puzzle is solved.

12. Customizing the Tile Prefab

The engine instantiates whatever Transform is assigned to the [Puzzle Tile Prefab](#) field.

Requirements for a custom prefab:

- **Collider tagged 'Tile':** The engine uses Physics ray-casting to detect taps. The collider must exist and carry the [Tile](#) tag.
- **Renderer with a Material:** The engine sets the main texture of the tile's material to the correct image slice at runtime.
- **PuzzleTile component:** The Puzzle Tile component must be present on the root of the prefab (or added automatically by the engine during instantiation).
- **Number text (optional):** A TextMeshPro component referenced by [PuzzleTile.numberText](#) is used to display the tile number. Remove it or leave the reference null if you don't want numbers.

Tip

The simplest approach is to duplicate the included tile prefab, modify its mesh or material, and assign the duplicate to the Puzzle Tile Prefab field.

13. Multiple Puzzle Managers in One Scene

Each Puzzle Manager instance is fully self-contained. You can place as many as you like in a scene and they will not interfere with each other.

Checklist for multiple managers

- Each manager has its own [Puzzle Container](#) transform.
- Each manager has its own [Puzzle Complete Image](#) object (or none).
- Each manager has its own dedicated [PuzzleSaveSO](#) asset assigned to its Save Slot.
- Each manager has its own [Puzzles List](#) (lists can share the same Puzzle assets if you want the same images to appear in multiple managers).

Note

Tile raycasting is collision-based. As long as tiles from different puzzle managers do not overlap physically, each manager will only respond to taps on its own tiles.

14. Frequently Asked Questions

- **The puzzle starts solved or obviously easy to solve.**

Check the [Scramble Count](#) on your Puzzle Scramble Difficulty asset. A very low value (e.g. 1 or 2) produces trivially easy starting positions. Try a value of 50 or higher for a well-mixed puzzle.

- **Tiles are not responding to clicks or taps.**

Make sure the tile prefab's Collider has the [Tile](#) tag assigned. Also confirm that the [Puzzle Container](#) field on the manager is assigned and that a camera tagged [MainCamera](#) exists in the scene.

- **Progress is not being saved between sessions.**

Check that a [PuzzleSaveSO](#) asset is assigned to the [Save Slot](#) field. If the field is empty, no saving occurs and a warning appears in the inspector.

- **I want to start fresh and ignore the existing save.**

Call [NewPuzzleAction\(true\)](#) to force a new puzzle and discard any saved state. You can also wire a UI button to this call.

- **Can I use non-square images?**

Yes. The engine slices and scales tiles to fit the puzzle container regardless of the source image's aspect ratio.

- **Can I have an uneven grid like 3 columns and 5 rows?**

Yes. Set [Slices Per Row](#) to 3 and [Slices Per Column](#) to 5 on your Puzzle Slice Difficulty asset. Any positive integer combination is valid.

- **I want the win SFX and animation but also need to run my own code on win.**

Both happen simultaneously. The engine handles the animation and SFX internally when the puzzle is solved, and it fires the [Win Function](#) Unity Event at the same time. Subscribe your own method to that event and it will run alongside the built-in behavior.

15. Quick Reference Card

Most Important Settings at a Glance

Setting	Type / Values	Description
Puzzles List	List<PuzzleSO>	The images available to this manager.
Puzzle Slice Difficulty	PuzzleSliceDifficultySO	Global grid size (rows x columns).
Puzzle Scramble Difficulty	PuzzleScrambleDifficultySO	How many moves are used to scramble.
Save Slot	PuzzleSaveSO	Leave empty to disable saving.
Puzzle Start Mode	Enum	LoadSave or StartNewPuzzle.
Save Interval Moves	int	0 = save on focus loss only.
Auto Start Puzzle	bool	Start on Awake, or call manually.
Win Function	UnityEvent	Fired when puzzle is solved.
Lose Function	UnityEvent	Fired when Lose() is called.

Key API Calls

Setting	Type / Values	Description
NewPuzzleAction(false)	method	Start or load a puzzle (respects Start Mode).
NewPuzzleAction(true)	method	Always start a fresh puzzle.
ReScrambleAction()	method	Re-scramble the current puzzle.
ShowTileNumberAction()	method	Toggle tile numbers on/off.
ShowCompletePuzzleImage(false)	method	Show the hint/solution image instantly.
ShowCompletePuzzleImage(true)	method	Show the hint/solution image with fade-in.
HideCompletePuzzleImage()	method	Return to the puzzle tiles.
Win()	method	Manually trigger the win state.

Setting	Type / Values	Description
Lose()	method	Manually trigger the lose state.
numberOfMoves	int	Current move count (read-only in practice).

Thank you for using 3D Sliding Puzzle Engine.
If you find this asset useful, please rate it on the Asset Store.

Thank you